

```
! pip install -U \
    langchain \
    langchain-core \
    langchain-community \
    langchain-groq \
    langchain-huggingface \
    sentence-transformers \
    faiss-cpu \
    python-dotenv
```

```
Requirement already satisfied: langchain in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (1.3.14)
Requirement already satisfied: langchain-core in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (1.5.3)
Requirement already satisfied: langchain-community in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (0.4.2)
Requirement already satisfied: langchain-groq in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (1.1.3)
Requirement already satisfied: langchain-huggingface in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (1.2.2)
Requirement already satisfied: sentence-transformers in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (5.6.1)
Requirement already satisfied: faiss-cpu in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (1.14.3)
Requirement already satisfied: python-dotenv in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (1.2.2)
Requirement already satisfied: langgraph<1.3.0,>=1.2.5 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langchain)
(1.2.9)
Requirement already satisfied: pydantic<3.0.0,>=2.7.4 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langchain)
(2.12.5)
Requirement already satisfied: jsonpatch<2.0.0,>=1.33.0 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langchain-
core) (1.33)
Requirement already satisfied: langchain-protocol>=0.0.17 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langchain-
core) (0.0.18)
Requirement already satisfied: langsmith<1.0.0,>=0.3.45 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langchain-
core) (0.10.10)
Requirement already satisfied: packaging>=23.2.0 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langchain-
core) (25.0)
Requirement already satisfied: pyyaml<7.0.0,>=5.3.0 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langchain-
core) (6.0.3)
Requirement already satisfied: tenacity!=8.4.0,<10.0.0,>=8.1.0 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langchain-
core) (9.1.2)
```

Requirement already satisfied: typing-extensions<5.0.0,>=4.7.0 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langchain-core) (4.15.0)

Requirement already satisfied: uuid-utils<1.0,>=0.12.0 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langchain-core) (0.17.0)

Requirement already satisfied: jsonpointer>=1.9 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from jsonpatch<2.0.0,>=1.33.0->langchain-core) (3.1.1)

Requirement already satisfied: langgraph-checkpoint<5.0.0,>=4.1.0 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langgraph<1.3.0,>=1.2.5->langchain) (4.1.1)

Requirement already satisfied: langgraph-prebuilt<1.2.0,>=1.1.0 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langgraph<1.3.0,>=1.2.5->langchain) (1.1.0)

Requirement already satisfied: langgraph-sdk<0.5.0,>=0.4.2 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langgraph<1.3.0,>=1.2.5->langchain) (0.4.2)

Requirement already satisfied: xxhash>=3.5.0 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langgraph<1.3.0,>=1.2.5->langchain) (3.8.1)

Requirement already satisfied: ormsgpack>=1.12.0 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langgraph-checkpoint<5.0.0,>=4.1.0->langgraph<1.3.0,>=1.2.5->langchain) (1.12.2)

Requirement already satisfied: httpx>=0.25.2 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langgraph-sdk<0.5.0,>=0.4.2->langgraph<1.3.0,>=1.2.5->langchain) (0.28.1)

Requirement already satisfied: orjson>=3.11.5 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langgraph-sdk<0.5.0,>=0.4.2->langgraph<1.3.0,>=1.2.5->langchain) (3.11.9)

Requirement already satisfied: websockets<16,>=14 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langgraph-sdk<0.5.0,>=0.4.2->langgraph<1.3.0,>=1.2.5->langchain) (15.0.1)

Requirement already satisfied: anyio>=3.5.0 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langsmith<1.0.0,>=0.3.45->langchain-core) (4.12.0)

Requirement already satisfied: distro>=1.7.0 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langsmith<1.0.0,>=0.3.45->langchain-core) (1.9.0)

Requirement already satisfied: requests-toolbelt>=1.0.0 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langsmith<1.0.0,>=0.3.45->langchain-core) (1.0.0)

Requirement already satisfied: requests>=2.0.0 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langsmith<1.0.0,>=0.3.45->langchain-core) (2.32.5)

Requirement already satisfied: sniffio>=1.1 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langsmith<1.0.0,>=0.3.45->langchain-core) (1.3.1)

Requirement already satisfied: zstandard>=0.23.0 in

```
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from
langsmith<1.0.0,>=0.3.45->langchain-core) (0.25.0)
Requirement already satisfied: certifi in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from
httpx>=0.25.2->langgraph-sdk<0.5.0,>=0.4.2->langgraph<1.3.0,>=1.2.5-
>langchain) (2025.11.12)
Requirement already satisfied: httpcore==1.* in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from
httpx>=0.25.2->langgraph-sdk<0.5.0,>=0.4.2->langgraph<1.3.0,>=1.2.5-
>langchain) (1.0.9)
Requirement already satisfied: idna in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from
httpx>=0.25.2->langgraph-sdk<0.5.0,>=0.4.2->langgraph<1.3.0,>=1.2.5-
>langchain) (3.11)
Requirement already satisfied: h11>=0.16 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from
httpcore==1.*->httpx>=0.25.2->langgraph-sdk<0.5.0,>=0.4.2-
>langgraph<1.3.0,>=1.2.5->langchain) (0.16.0)
Requirement already satisfied: annotated-types>=0.6.0 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from
pydantic<3.0.0,>=2.7.4->langchain) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.5 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from
pydantic<3.0.0,>=2.7.4->langchain) (2.41.5)
Requirement already satisfied: typing-inspection>=0.4.2 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from
pydantic<3.0.0,>=2.7.4->langchain) (0.4.2)
Requirement already satisfied: aiohttp<4.0.0,>=3.8.3 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langchain-
community) (3.14.3)
Requirement already satisfied: httpx-sse<1.0.0,>=0.4.0 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langchain-
community) (0.4.3)
Requirement already satisfied: langchain-classic<2.0.0,>=1.0.7 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langchain-
community) (1.0.8)
Requirement already satisfied: numpy>=1.26.2 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langchain-
community) (2.3.0)
Requirement already satisfied: pydantic-settings<3.0.0,>=2.10.1 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langchain-
community) (2.14.2)
Requirement already satisfied: sqlalchemy<3.0.0,>=1.4.0 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langchain-
community) (2.0.51)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from
aiohttp<4.0.0,>=3.8.3->langchain-community) (2.7.1)
Requirement already satisfied: aiosignal>=1.4.0 in
```

/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from aiohttp<4.0.0,>=3.8.3->langchain-community) (1.4.0)
Requirement already satisfied: attrs>=17.3.0 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from aiohttp<4.0.0,>=3.8.3->langchain-community) (25.4.0)
Requirement already satisfied: frozenlist>=1.1.1 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from aiohttp<4.0.0,>=3.8.3->langchain-community) (1.8.0)
Requirement already satisfied: multidict<7.0,>=4.5 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from aiohttp<4.0.0,>=3.8.3->langchain-community) (6.7.1)
Requirement already satisfied: propcache>=0.2.0 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from aiohttp<4.0.0,>=3.8.3->langchain-community) (0.5.2)
Requirement already satisfied: yarl<2.0,>=1.17.0 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from aiohttp<4.0.0,>=3.8.3->langchain-community) (1.24.5)
Requirement already satisfied: langchain-text-splitters<2.0.0,>=1.1.2 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langchain-classic<2.0.0,>=1.0.7->langchain-community) (1.1.2)
Requirement already satisfied: charset_normalizer<4,>=2 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from requests>=2.0.0->langsmith<1.0.0,>=0.3.45->langchain-core) (3.4.4)
Requirement already satisfied: urllib3<3,>=1.21.1 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from requests>=2.0.0->langsmith<1.0.0,>=0.3.45->langchain-core) (2.6.2)
Requirement already satisfied: groq<1.0.0,>=0.30.0 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langchain-groq) (0.37.1)
Requirement already satisfied: huggingface-hub<2.0.0,>=0.33.4 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langchain-huggingface) (1.25.1)
Requirement already satisfied: tokenizers<1.0.0,>=0.19.1 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from langchain-huggingface) (0.22.2)
Requirement already satisfied: click<9.0.0,>=8.4.2 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from huggingface-hub<2.0.0,>=0.33.4->langchain-huggingface) (8.4.2)
Requirement already satisfied: filelock>=3.10.0 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from huggingface-hub<2.0.0,>=0.33.4->langchain-huggingface) (3.32.0)
Requirement already satisfied: fsspec>=2023.5.0 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from huggingface-hub<2.0.0,>=0.33.4->langchain-huggingface) (2026.6.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.5.1 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from huggingface-hub<2.0.0,>=0.33.4->langchain-huggingface) (1.5.2)
Requirement already satisfied: tqdm>=4.42.1 in /opt/anaconda3/envs/jub/lib/python3.12/site-packages (from

huggingface-hub<2.0.0,>=0.33.4->langchain-huggingface) (4.67.1)
Requirement already satisfied: transformers<6.0.0,>=4.41.0 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from sentence-
transformers) (5.14.1)
Requirement already satisfied: torch>=1.11.0 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from sentence-
transformers) (2.13.0)
Requirement already satisfied: scikit-learn>=0.22.0 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from sentence-
transformers) (1.8.0)
Requirement already satisfied: scipy>=1.0.0 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from sentence-
transformers) (1.16.3)
Requirement already satisfied: regex>=2025.10.22 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from
transformers<6.0.0,>=4.41.0->sentence-transformers) (2026.7.19)
Requirement already satisfied: typer in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from
transformers<6.0.0,>=4.41.0->sentence-transformers) (0.27.0)
Requirement already satisfied: safetensors>=0.8.0 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from
transformers<6.0.0,>=4.41.0->sentence-transformers) (0.8.0)
Requirement already satisfied: joblib>=1.3.0 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from scikit-
learn>=0.22.0->sentence-transformers) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.2.0 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from scikit-
learn>=0.22.0->sentence-transformers) (3.6.0)
Requirement already satisfied: setuptools>=77.0.3 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from
torch>=1.11.0->sentence-transformers) (80.9.0)
Requirement already satisfied: sympy>=1.13.3 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from
torch>=1.11.0->sentence-transformers) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from
torch>=1.11.0->sentence-transformers) (3.6.1)
Requirement already satisfied: jinja2 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from
torch>=1.11.0->sentence-transformers) (3.1.6)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from
sympy>=1.13.3->torch>=1.11.0->sentence-transformers) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from jinja2-
>torch>=1.11.0->sentence-transformers) (3.0.3)
Requirement already satisfied: shellingham>=1.3.0 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from typer-
>transformers<6.0.0,>=4.41.0->sentence-transformers) (1.5.4)

Requirement already satisfied: rich>=13.8.0 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from typer->transformers<6.0.0,>=4.41.0->sentence-transformers) (15.0.0)
Requirement already satisfied: annotated-doc>=0.0.2 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from typer->transformers<6.0.0,>=4.41.0->sentence-transformers) (0.0.4)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from rich>=13.8.0->typer->transformers<6.0.0,>=4.41.0->sentence-transformers) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from rich>=13.8.0->typer->transformers<6.0.0,>=4.41.0->sentence-transformers) (2.19.2)
Requirement already satisfied: mdurl~=0.1 in
/opt/anaconda3/envs/jub/lib/python3.12/site-packages (from markdown-it-py>=2.2.0->rich>=13.8.0->typer->transformers<6.0.0,>=4.41.0->sentence-transformers) (0.1.2)

```
import os
```

```
from dotenv import load_dotenv
```

```
from langchain_core.documents import Document  
from langchain_core.output_parsers import StrOutputParser  
from langchain_core.prompts import ChatPromptTemplate  
from langchain_core.runnables import RunnablePassthrough
```

```
from langchain_community.vectorstores import FAISS  
from langchain_groq import ChatGroq  
from langchain_huggingface import HuggingFaceEmbeddings
```

```
/opt/anaconda3/envs/jub/lib/python3.12/site-packages/tqdm/auto.py:21:  
TqdmWarning: IProgress not found. Please update jupyter and  
ipywidgets. See
```

```
https://ipywidgets.readthedocs.io/en/stable/user\_install.html
```

```
    from .autonotebook import tqdm as notebook_tqdm  
/var/folders/9f/00d3gqls0w3_0m391myx76400000gn/T/ipykernel_1679/119050  
6230.py:10: DeprecationWarning: `langchain-community` is being sunset  
and is no longer actively maintained. See  
https://github.com/langchain-ai/langchain-community/issues/674 for  
details and migration guidance toward standalone integration packages.
```

```
    from langchain_community.vectorstores import FAISS
```

```
import warnings
```

```
# Suppress all warnings  
warnings.filterwarnings("ignore")
```

```
# --- YOUR CODE HERE ---
```

```

# Load the variables from the .env file into the environment
load_dotenv()

#2 Access your key securely
api_key = os.getenv('GROQ_API_KEY')

documents = [
    Document(
        page_content=(
            "LangGraph is a framework for building stateful "
            "and multi-agent AI applications."
        ),
        metadata={"source": "langgraph_notes"},
    ),
    Document(
        page_content=(
            "Inception BD is a Edtech Platform "
            "and provides courses on AI."
        ),
        metadata={"source": "inception_notes"},
    ),
    Document(
        page_content=(
            "RAG stands for Retrieval-Augmented Generation. "
            "It retrieves relevant information before generating an
answer."
        ),
        metadata={"source": "rag_notes"},
    ),
    Document(
        page_content=(
            "Groq provides fast inference for supported large "
            "language models through the Groq API."
        ),
        metadata={"source": "groq_notes"},
    ),
    Document(
        page_content=(
            "FAISS is a vector similarity-search library. "
            "It can retrieve documents whose embeddings are close "
            "to the query embedding."
        ),
        metadata={"source": "faiss_notes"},
    ),
    Document(
        page_content=(
            "Jubayer is AI Enginner "
            "He is working"

```

```

        ),
        metadata={'source': "Jubayer"}
    )
]

documents

[Document(metadata={'source': 'langgraph_notes'},
page_content='LangGraph is a framework for building stateful and
multi-agent AI applications.'),
 Document(metadata={'source': 'inception_notes'},
page_content='Inception BD is a Edtech PPlatform and provides courses
on AI.'),
 Document(metadata={'source': 'rag_notes'}, page_content='RAG stands
for Retrieval-Augmented Generation. It retrieves relevant information
before generating an answer.'),
 Document(metadata={'source': 'groq_notes'}, page_content='Groq
provides fast inference for supported large language models through
the Groq API.'),
 Document(metadata={'source': 'faiss_notes'}, page_content='FAISS is a
vector similarity-search library. It can retrieve documents whose
embeddings are close to the query embedding.')]

type(documents[0])

langchain_core.documents.base.Document

```

embedding mode all minillmv-6

```

embeddings = HuggingFaceEmbeddings(
    model_name = "sentence-transformers/all-MiniLM-L6-v2",
    encode_kwargs = {
        "normalize_embeddings": True,
    },
)

Loading weights: 100%|██████████| 103/103 [00:00<00:00, 7938.21it/s]

```

Create the FAISS Vector store and store embeddings

```

vector_store = FAISS.from_documents(
    documents=documents,
    embedding=embeddings,
)

```

save to the local load local

```

vector_store.save_local('faiss_index')

```



```
vector_store = FAISS.load_local('lfaiss_index', embeddings,
allow_dangerous_deserialization=True)
```

Crate Retriver

vetor is knowlaged bas

```
retriver = vector_store.as_retriever(
    search_type = 'similarity',
    search_kwargs = {"k": 2}# k = 2 two result present or k=3 result
present
)
```

LLM mode l connection

```
load_dotenv()
api_key = os.getenv("GROQ_API_KEY")

llm = ChatGroq(
    model= 'llama-3.1-8b-instant',
    temperature=0, # not be creative 0, use to be knowledge 1 is
crativeat
    max_retries=2, # 2 try to connect two times
    api_key=
'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'
)

response =llm.invoke("Hi tell me about you google ")

response.content

'I\'m not Google, but I\'m a large language model (LLM) provided by
Google. I\'m an AI designed to assist and communicate with users in a
helpful and informative way.\n\nHere\'s a brief overview of Google:\n\n
**What is Google?**\n\nGoogle is a multinational technology company
that specializes in Internet-related services and products. It was
founded in 1998 by Larry Page and Sergey Brin while they were Ph.D.
students at Stanford University in California.\n\n**Google\'s Products
and Services**\n\nGoogle offers a wide range of products and services,
including:\n\n1. **Search Engine**: Google\'s search engine is the
most widely used search engine in the world, allowing users to search
for information on the web.\n2. **Google Ads**: A platform for
businesses to create and display ads on Google\'s search engine and
other websites.\n3. **Google Maps**: A mapping service that provides
directions, satellite imagery, and street views of locations around
the world.\n4. **YouTube**: A video-sharing platform where users can
upload, share, and view videos.\n5. **Google Drive**: A cloud storage
service that allows users to store and access files from anywhere.\n6.
**Google Docs, Sheets, and Slides**: A suite of productivity software
that allows users to create and edit documents, spreadsheets, and
```

presentations online.\n7. **Google Chrome**: A web browser that is widely used for browsing the internet.\n8. **Google Assistant**: A virtual assistant that can perform tasks, answer questions, and control smart home devices.\n9. **Google Home**: A smart speaker that allows users to control their smart home devices and access information using voice commands.\n\n**Google's Mission**\n\nGoogle's mission is to "organize the world's information and make it universally accessible and useful." The company aims to provide accurate and relevant information to users, while also promoting innovation and progress in the fields of technology and science.\n\n**Google's Impact**\n\nGoogle has had a significant impact on the world, transforming the way people access and use information. The company's products and services have become an integral part of modern life, and its influence can be seen in many areas, including:\n\n1. **E-commerce**: Google's search engine and advertising platform have revolutionized the way people shop online.\n2. **Education**: Google's products and services have made it easier for people to access information and learn new skills.\n3. **Communication**: Google's messaging apps, such as Gmail and Google Hangouts, have changed the way people communicate with each other.\n4. **Healthcare**: Google's products and services have improved access to healthcare information and services.\n\nOverall, Google is a technology company that has had a profound impact on the world, and its products and services continue to shape the way people live, work, and interact with each other.'

```
response = llm.invoke("tell me about uk")
```

```
response.content
```

'The United Kingdom (UK) is a sovereign state located in Northwest Europe, comprising four constituent countries: England, Scotland, Wales, and Northern Ireland. Here's an overview of the UK:\n\n**Geography and Climate**\n\nThe UK is an archipelago of over 6,000 islands, with the largest being Great Britain (England, Scotland, and Wales) and Ireland (Northern Ireland). The UK shares a border with the Republic of Ireland to the west. The country's terrain is diverse, with mountains, hills, and coastal plains. The climate is generally temperate, with mild winters and cool summers.\n\n**History**\n\nThe UK has a rich and complex history, with evidence of human habitation dating back to the Mesolithic era. The Romans conquered the island in 43 AD, followed by the Anglo-Saxons, Vikings, and Normans. The UK became a unified state in 1707, with the Acts of Union merging England and Scotland. The Industrial Revolution transformed the country's economy, and the UK became a major world power.\n\n**Government and Politics**\n\nThe UK is a constitutional monarchy, with a parliamentary system of government. The monarch (currently King Charles III) serves as head of state, while the Prime Minister (currently Rishi Sunak) is the head of government. The UK Parliament is bicameral, consisting of the House of Commons and the House of

Lords.\n\n**Economy**\n\nThe UK has a mixed economy, with a strong service sector, a significant manufacturing industry, and a growing digital economy. The country is a member of the G7 and the G20, and its economy is one of the largest in the world. The UK is also a significant player in international trade, with a strong focus on services, finance, and technology.\n\n**Culture**\n\nThe UK is known for its rich cultural heritage, with a diverse range of arts, music, literature, and cuisine. The country is home to many world-class museums, galleries, and theaters, including the British Museum, the National Gallery, and the Royal Shakespeare Company. The UK is also famous for its music scene, with iconic bands like The Beatles, The Rolling Stones, and Queen.\n\n**Education**\n\nThe UK has a highly regarded education system, with many world-class universities, including Oxford and Cambridge. The country is home to a diverse range of educational institutions, from primary schools to universities, and is a popular destination for international students.\n\n**Tourism**\n\nThe UK is a popular tourist destination, with millions of visitors each year. The country is home to many iconic landmarks, including Buckingham Palace, the Tower of London, and Stonehenge. The UK is also known for its beautiful countryside, with many national parks and areas of outstanding natural beauty.\n\n**Key Facts**\n\n* Capital city: London\n* Population: approximately 67 million\n* Language: English (official), Welsh, Scottish Gaelic, and Northern Irish dialects\n* Currency: Pound sterling (GBP)\n* Time zone: Greenwich Mean Time (GMT)\n* National anthem: "God Save the King"\n\n**Interesting Facts**\n\n* The UK is home to the oldest university in the English-speaking world, the University of Oxford, which was founded in 1167.\n* The UK is the birthplace of the Industrial Revolution, with the first steam engine invented by James Watt in 1769.\n* The UK is home to the world's oldest known pub, the Ye Olde Trip to Jerusalem, which dates back to 1189.\n* The UK is a leader in the field of science, with many world-class research institutions and scientists, including Stephen Hawking and Isaac Newton.\n\nI hope this gives you a good overview of the UK!'

Create the prompt

```
prompt = ChatPromptTemplate.from_template(
    """
    You are a helpful assistant.

    Answer the question using only the provided context.

    If the answer is not present in the context, say:
    "I do not know based on the provided context."

    Context:
    {context}

    Question:
```

```

        {question}

        Answer:
        """
    )

```

Format retrieved document

```

def format_documents(retrieved_documents: list[Document]) -> str:
    return "\n\n".join(
        document.page_content
        for document in retrieved_documents
    )

```

Build the RAG chain

```

rag_chain = (
    {
        "context": retriever | format_documents,
        'question': RunnablePassthrough(),
    }
    | prompt
    | llm
    | StrOutputParser() # make nice out put
)

```

Ask question

```

def ask_question(question: str) -> str:
    if not question.strip():
        raise ValueError('Question cannot be empty. ')

    return rag_chain.invoke(question)

if __name__ == "__main__":
    while True:
        user_question = input(
            "\nAsk a question of type 'text' : "
        ).strip()

        if user_question.lower() == 'exit':
            print("Application closed ")
            break

        try:
            answer = ask_question(user_question)

            print("\nAnswer :")
            print(answer)

```

```
except Exception as error:  
    print(f"\nError : {error}")
```

Answer :
Jubayer is AI Engineer.

Answer :
RAG stands for Retrieval-Augmented Generation. It retrieves relevant information before generating an answer.

Answer :
Inception BD is a Edtech Platform and provides courses on AI.
Application closed